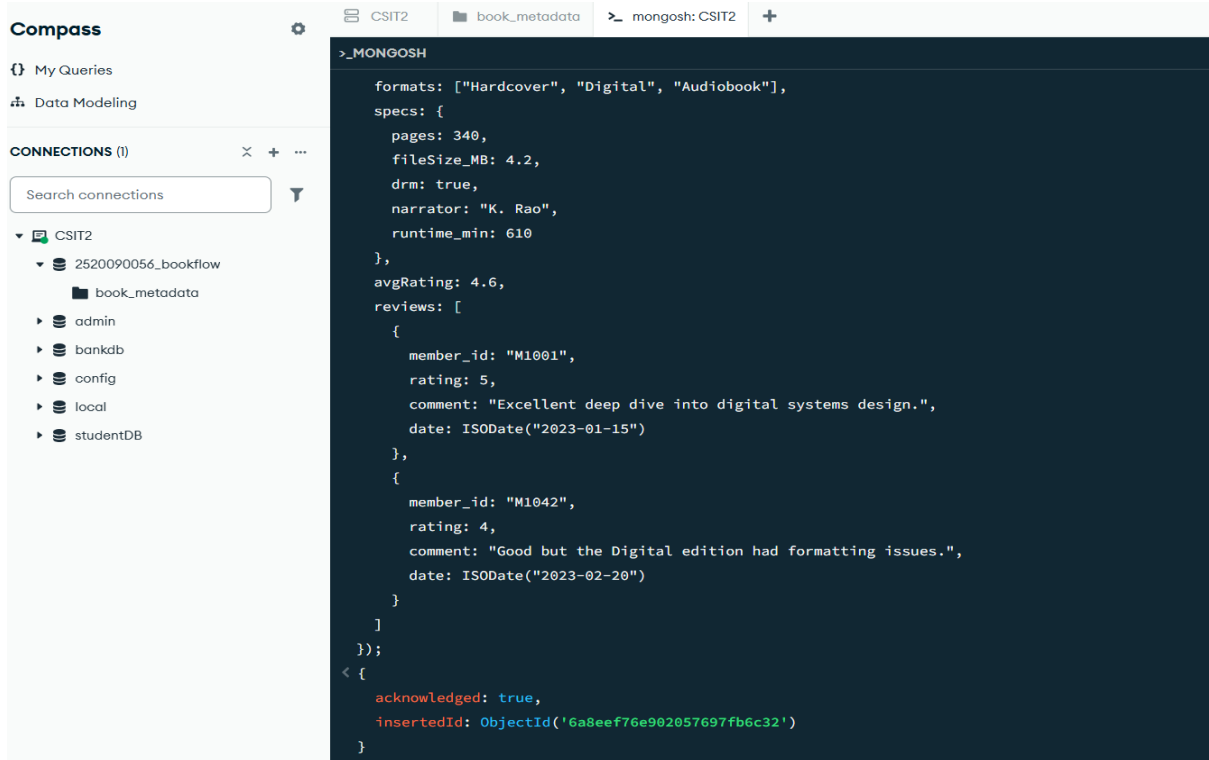


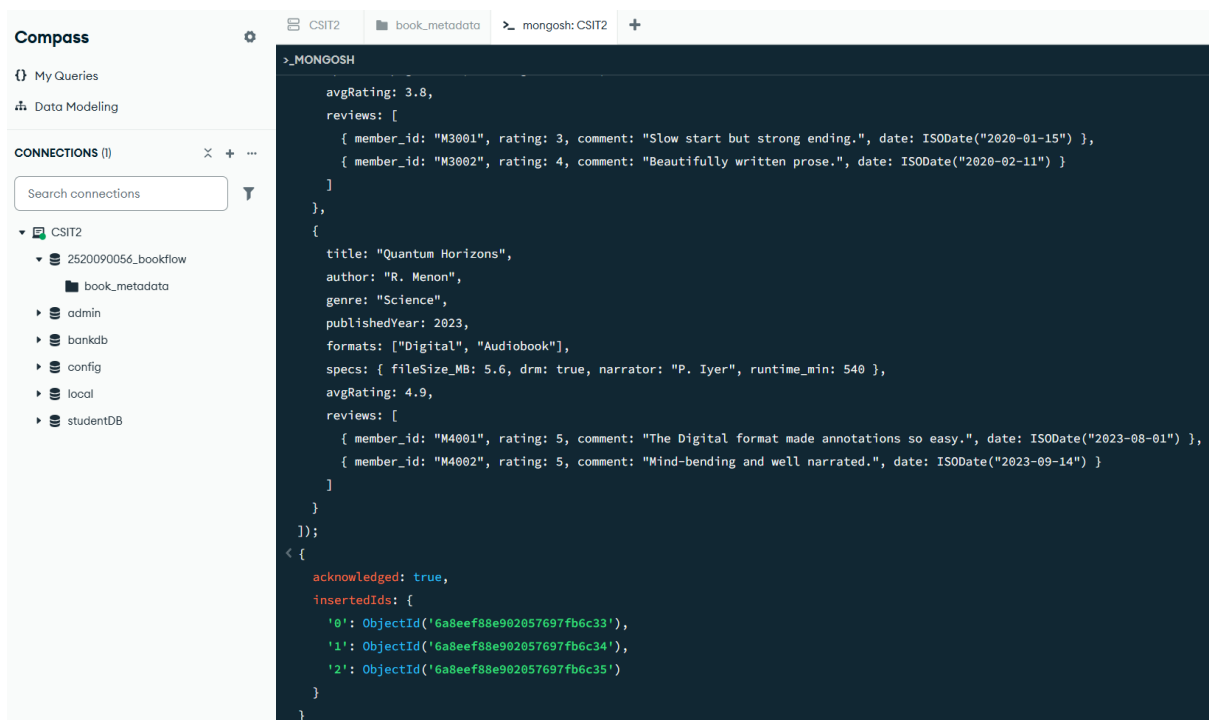
Task -1: Document Modelling



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'CSIT2' database, which contains a '2520090056_bookflow' collection. The 'book_metadata' collection is selected. The main panel displays a document in the 'book_metadata' collection. The document has the following fields: 'formats' (an array of 'Hardcover', 'Digital', and 'Audiobook'), 'specs' (an object with 'pages: 340', 'fileSize_MB: 4.2', 'drm: true', 'narrator: "K. Rao"', and 'runtime_min: 610'), 'avgRating: 4.6', and 'reviews' (an array of two review objects). The first review is from member 'M1001' with a rating of 5 and a comment 'Excellent deep dive into digital systems design.' dated '2023-01-15'. The second review is from member 'M1042' with a rating of 4 and a comment 'Good but the Digital edition had formatting issues.' dated '2023-02-20'. The document is acknowledged and has an inserted ID of '6a8eef76e902057697fb6c32'.

```
>_MONGOSH

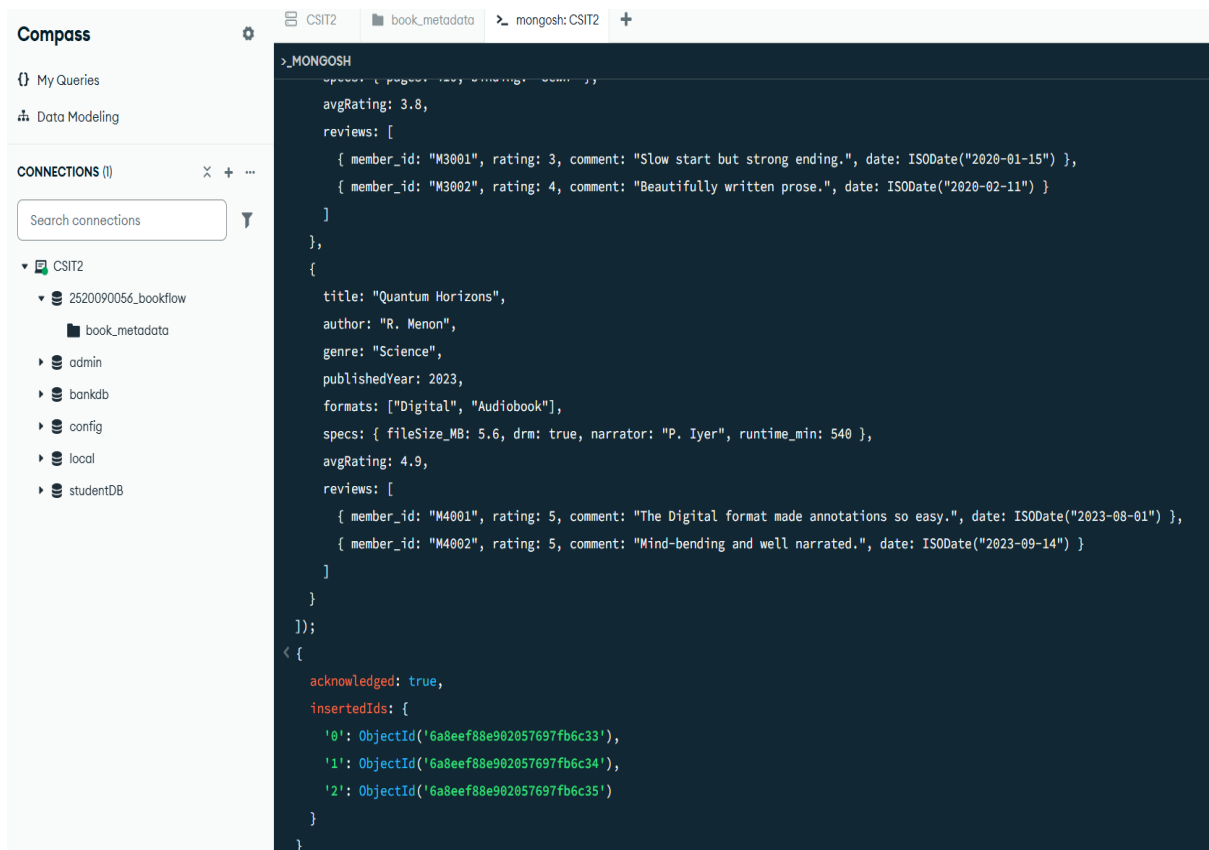
formats: ["Hardcover", "Digital", "Audiobook"],
specs: {
  pages: 340,
  fileSize_MB: 4.2,
  drm: true,
  narrator: "K. Rao",
  runtime_min: 610
},
avgRating: 4.6,
reviews: [
  {
    member_id: "M1001",
    rating: 5,
    comment: "Excellent deep dive into digital systems design.",
    date: ISODate("2023-01-15")
  },
  {
    member_id: "M1042",
    rating: 4,
    comment: "Good but the Digital edition had formatting issues.",
    date: ISODate("2023-02-20")
  }
]
});
< {
  acknowledged: true,
  insertedId: ObjectId('6a8eef76e902057697fb6c32')
}
```



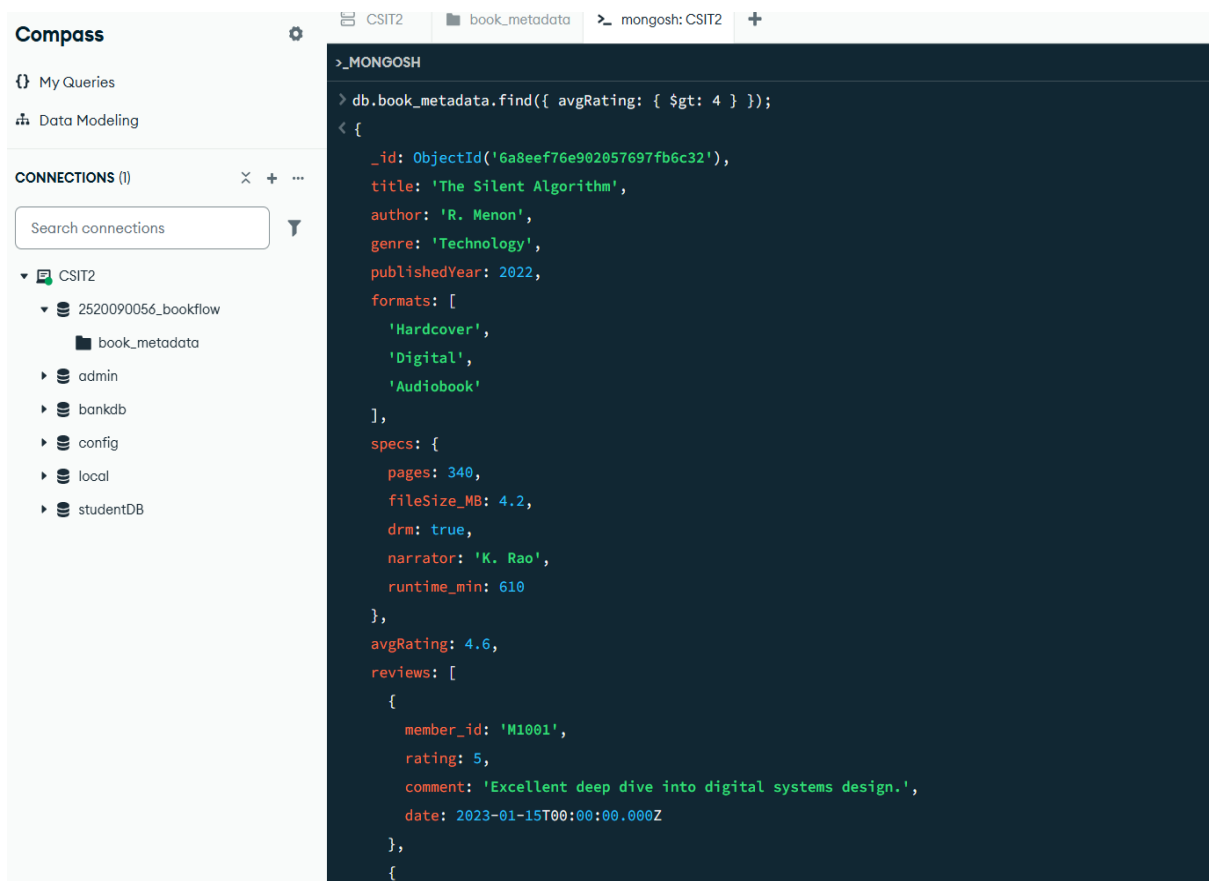
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'CSIT2' database, which contains a '2520090056_bookflow' collection. The 'book_metadata' collection is selected. The main panel displays a document in the 'book_metadata' collection. The document has the following fields: 'avgRating: 3.8', 'reviews' (an array of two review objects), 'title: "Quantum Horizons"', 'author: "R. Menon"', 'genre: "Science"', 'publishedYear: 2023', 'formats' (an array of 'Digital' and 'Audiobook'), 'specs' (an object with 'fileSize_MB: 5.6', 'drm: true', 'narrator: "P. Iyer"', and 'runtime_min: 540'), and 'avgRating: 4.9'. The first review is from member 'M3001' with a rating of 3 and a comment 'Slow start but strong ending.' dated '2020-01-15'. The second review is from member 'M3002' with a rating of 4 and a comment 'Beautifully written prose.' dated '2020-02-11'. The document is acknowledged and has three inserted IDs: '6a8eef88e902057697fb6c33', '6a8eef88e902057697fb6c34', and '6a8eef88e902057697fb6c35'.

```
>_MONGOSH

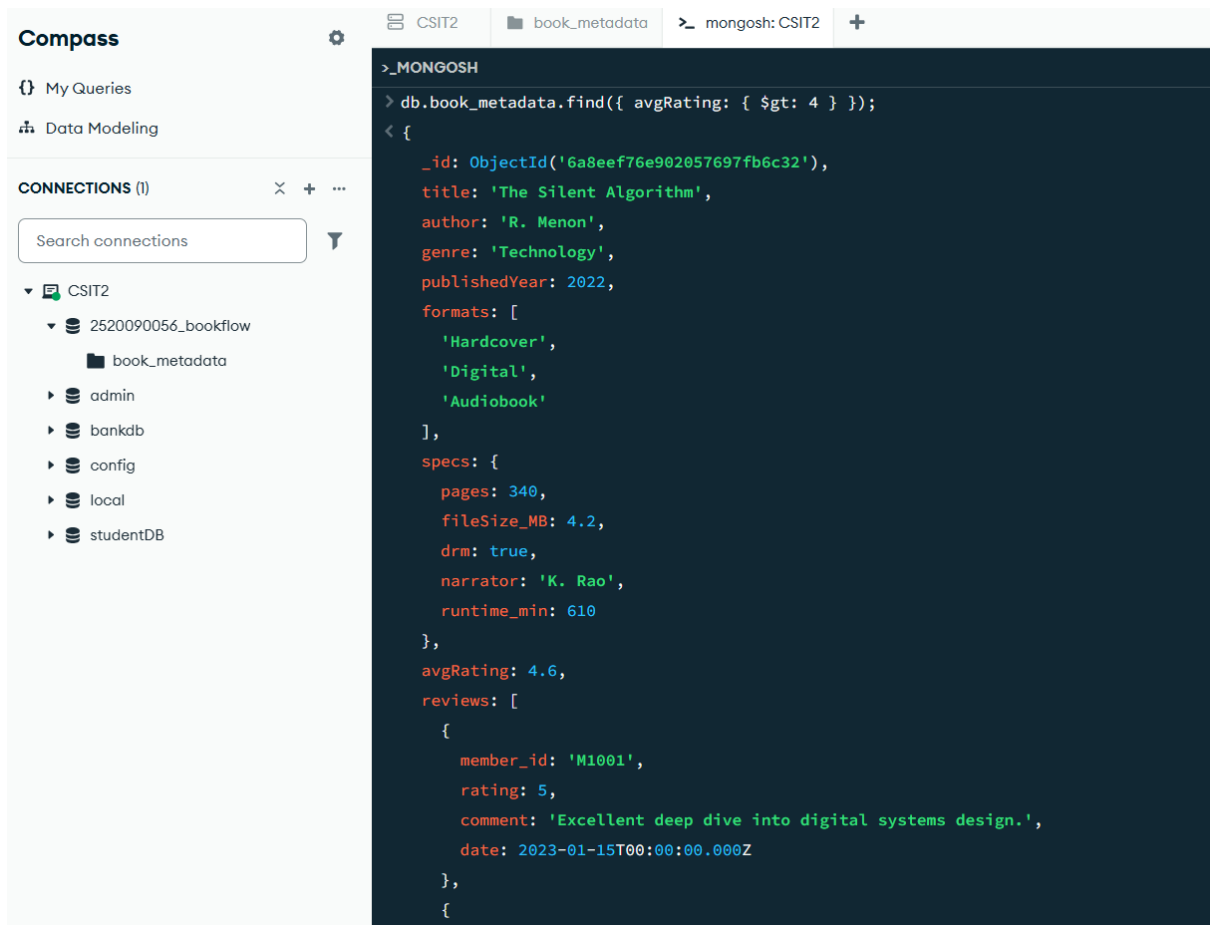
avgRating: 3.8,
reviews: [
  { member_id: "M3001", rating: 3, comment: "Slow start but strong ending.", date: ISODate("2020-01-15") },
  { member_id: "M3002", rating: 4, comment: "Beautifully written prose.", date: ISODate("2020-02-11") }
],
{
  title: "Quantum Horizons",
  author: "R. Menon",
  genre: "Science",
  publishedYear: 2023,
  formats: ["Digital", "Audiobook"],
  specs: { fileSize_MB: 5.6, drm: true, narrator: "P. Iyer", runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    { member_id: "M4001", rating: 5, comment: "The Digital format made annotations so easy.", date: ISODate("2023-08-01") },
    { member_id: "M4002", rating: 5, comment: "Mind-bending and well narrated.", date: ISODate("2023-09-14") }
  ]
}
});
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8eef88e902057697fb6c33'),
    '1': ObjectId('6a8eef88e902057697fb6c34'),
    '2': ObjectId('6a8eef88e902057697fb6c35')
  }
}
```



Task 2: Advancing query



1. Books rated above 4 (\$gt)



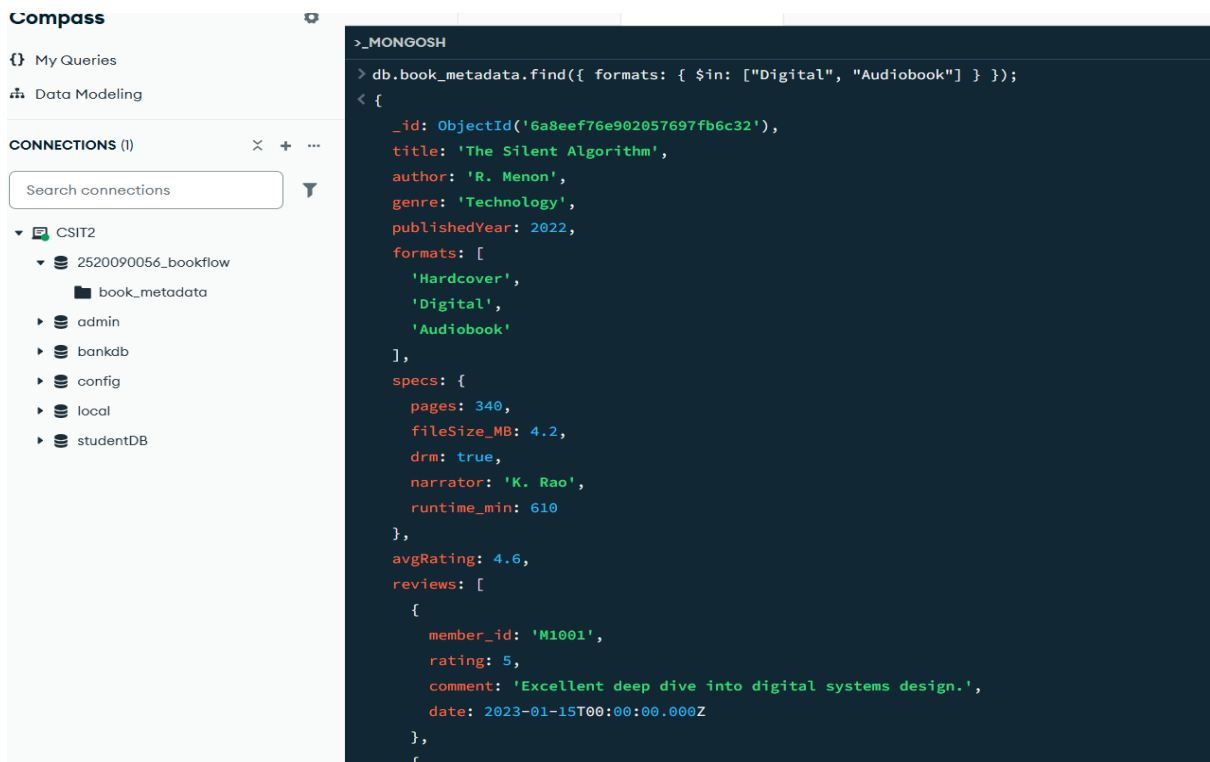
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel lists the 'CSIT2' database, with the 'book_metadata' collection selected. The main editor on the right shows a MongoDB query in the '>_MONGOSH' tab:

```
> db.book_metadata.find({ avgRating: { $gt: 4 } });
```

The result is a single document:

```
< {
  _id: ObjectId('6a8eef76e902057697fb6c32'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ],
}
```

2. Books available in specific formats (\$in)



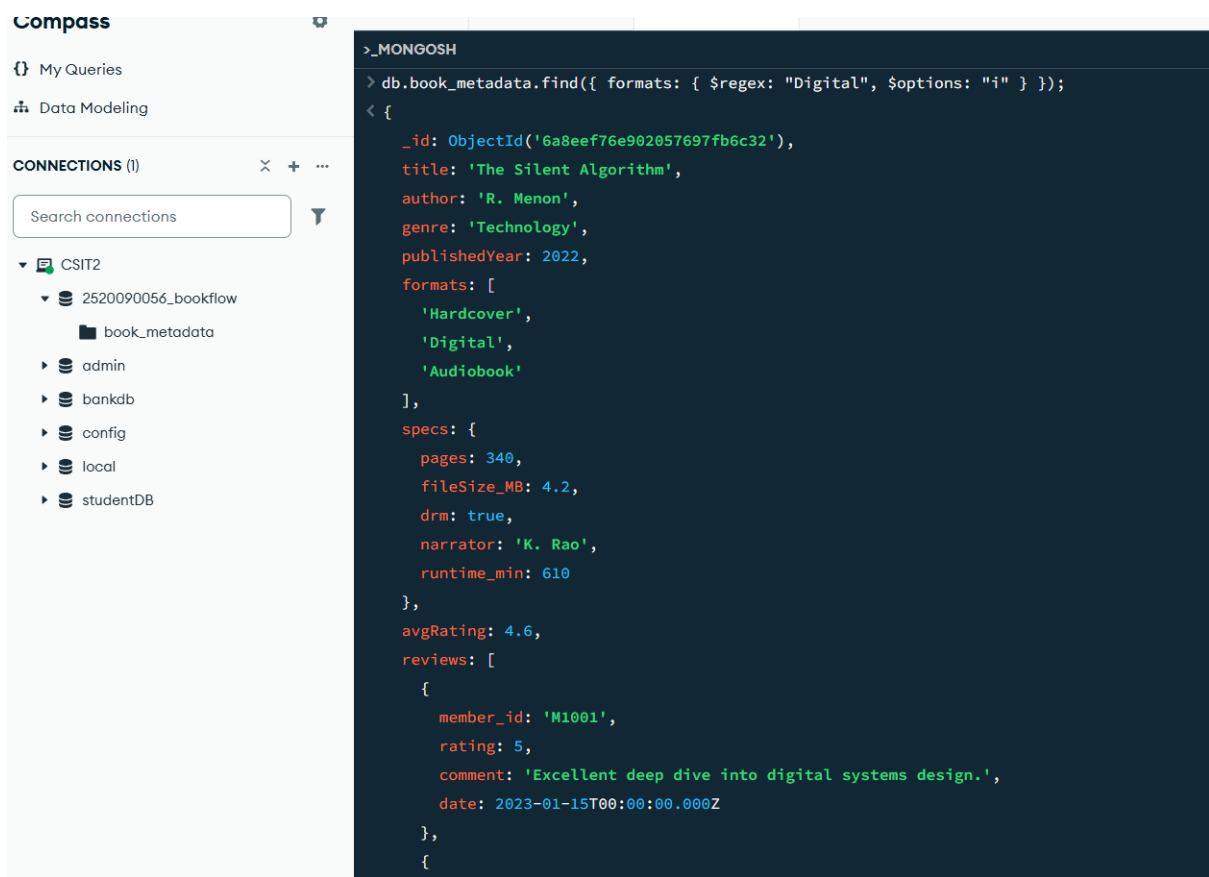
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel lists the 'CSIT2' database, with the 'book_metadata' collection selected. The main editor on the right shows a MongoDB query in the '>_MONGOSH' tab:

```
> db.book_metadata.find({ formats: { $in: ["Digital", "Audiobook"] } });
```

The result is a single document:

```
< {
  _id: ObjectId('6a8eef76e902057697fb6c32'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ],
}
```

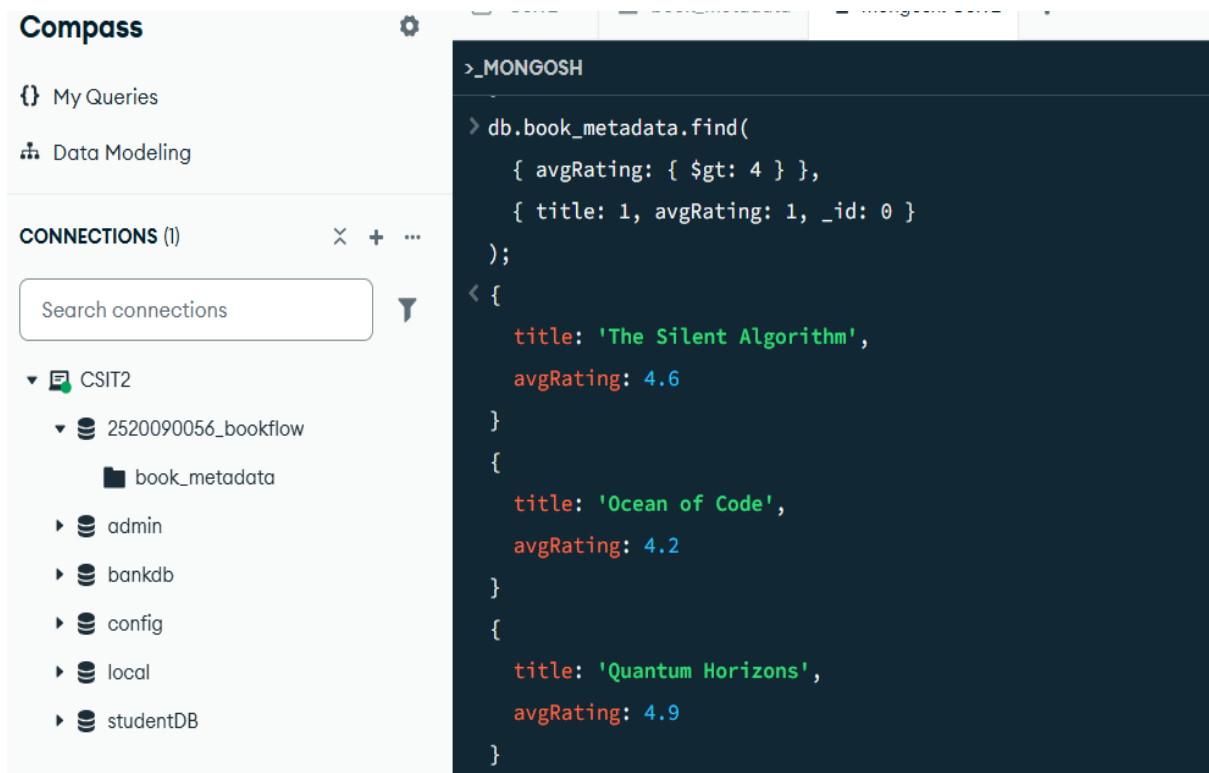
3. Combine both with \$and (implicit)



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists a connection named 'CSIT2' with a database '2520090056_bookflow' and a collection 'book_metadata'. The main panel displays a MongoDB query in the '_MONGOSH' editor:

```
>_MONGOSH
> db.book_metadata.find({ formats: { $regex: "Digital", $options: "i" } });
< {
  _id: ObjectId('6a8eef76e902057697fb6c32'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ],
}
```

4. Regex query — books with "Digital" format



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists a connection named 'CSIT2' with a database '2520090056_bookflow' and a collection 'book_metadata'. The main panel displays a MongoDB query in the '_MONGOSH' editor:

```
>_MONGOSH
> db.book_metadata.find(
  { avgRating: { $gt: 4 } },
  { title: 1, avgRating: 1, _id: 0 }
);
< {
  title: 'The Silent Algorithm',
  avgRating: 4.6
}
{
  title: 'Ocean of Code',
  avgRating: 4.2
}
{
  title: 'Quantum Horizons',
  avgRating: 4.9
}
```

5. Regex query — reviews mentioning "Digital" in comments (nested field)

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'CSIT2' database, which contains a collection named 'book_metadata'. The main panel displays a MongoDB query in the '>_MONGOSH' terminal:

```
>_MONGOSH
> db.book_metadata.find(
  { "reviews.comment": { $regex: "Digital", $options: "i" } },
  { title: 1, "reviews.$": 1 }
);
```

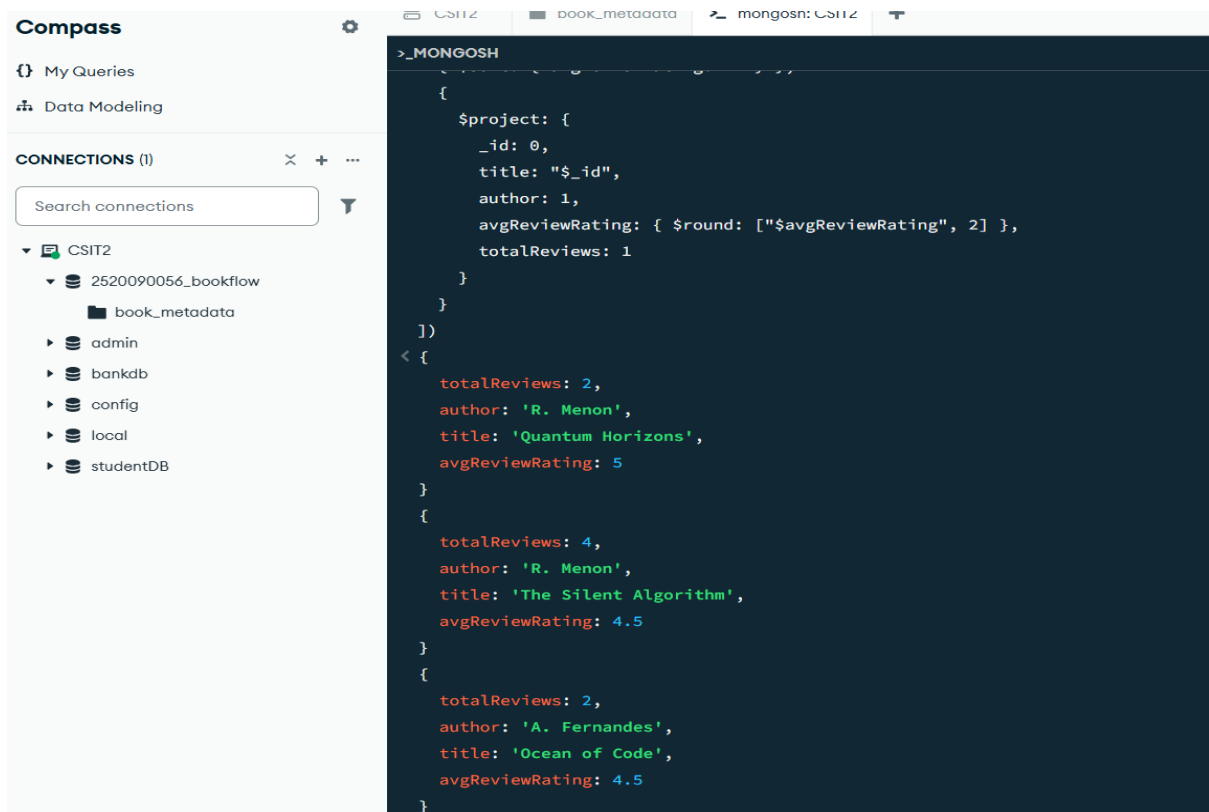
The query returns two documents from the 'book_metadata' collection:

```
< {
  _id: ObjectId('6a8eef76e902057697fb6c32'),
  title: 'The Silent Algorithm',
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ]
}
{
  _id: ObjectId('6a8eef88e902057697fb6c33'),
  title: 'Ocean of Code',
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: 2022-05-10T00:00:00.000Z
    }
  ]
}
```

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'CSIT2' database, which contains a collection named 'book_metadata'. The main panel displays a MongoDB query in the '>_MONGOSH' terminal:

```
>_MONGOSH
{
  _id: ObjectId('6a8eef88e902057697fb6c35'),
  title: 'Quantum Horizons',
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: 2023-08-01T00:00:00.000Z
    }
  ]
}
{
  _id: ObjectId('6a8ef0aa25577830d9d7c4cb'),
  title: 'The Silent Algorithm',
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: '2023-01-15'
    }
  ]
}
```

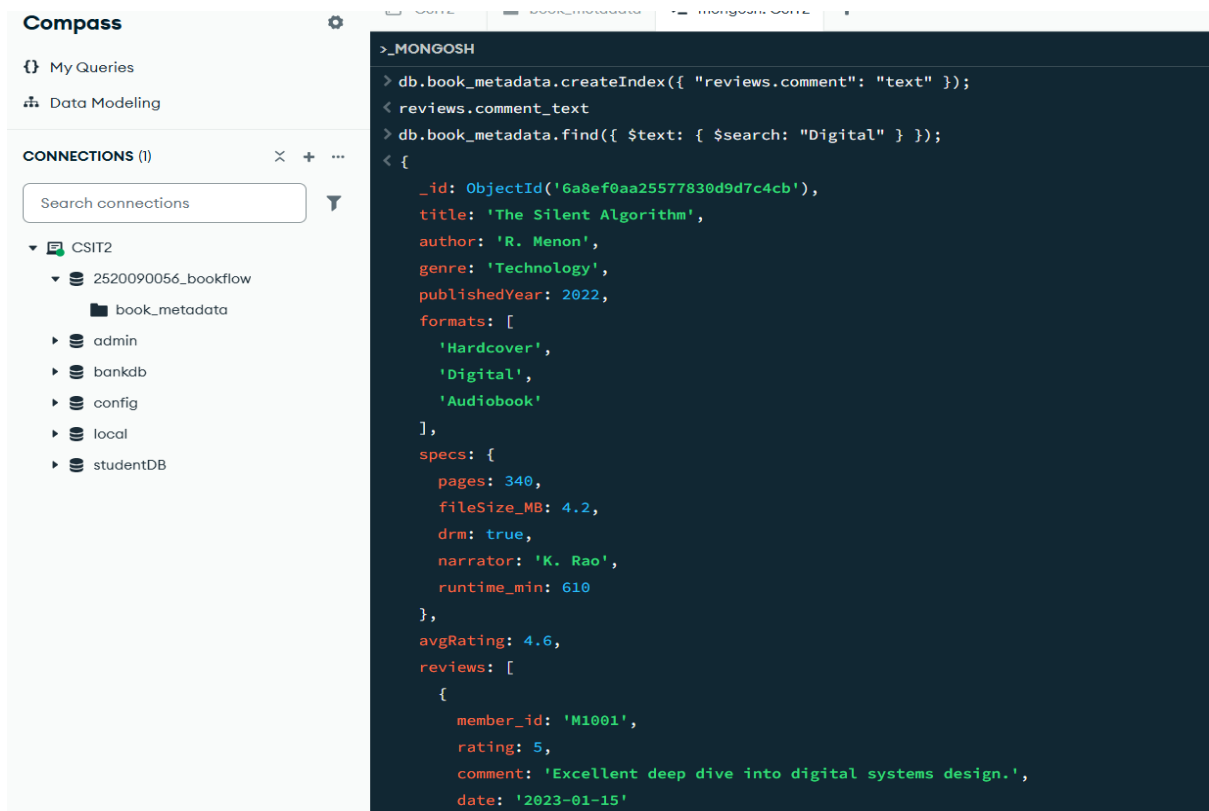
Task 3: Aggregation Pipeline — "Top Rated" Report



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists a connection named 'CSIT2' with a database '2520090056_bookflow' and a collection 'book_metadata'. The main panel displays the MONGOSH console with the following aggregation pipeline:

```
>_MONGOSH
{
  $project: {
    _id: 0,
    title: "$_id",
    author: 1,
    avgReviewRating: { $round: ["$avgReviewRating", 2] },
    totalReviews: 1
  }
}
))
< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'Quantum Horizons',
  avgReviewRating: 5
}
{
  totalReviews: 4,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
}
{
  totalReviews: 2,
  author: 'A. Fernandes',
  title: 'Ocean of Code',
  avgReviewRating: 4.5
}
}
```

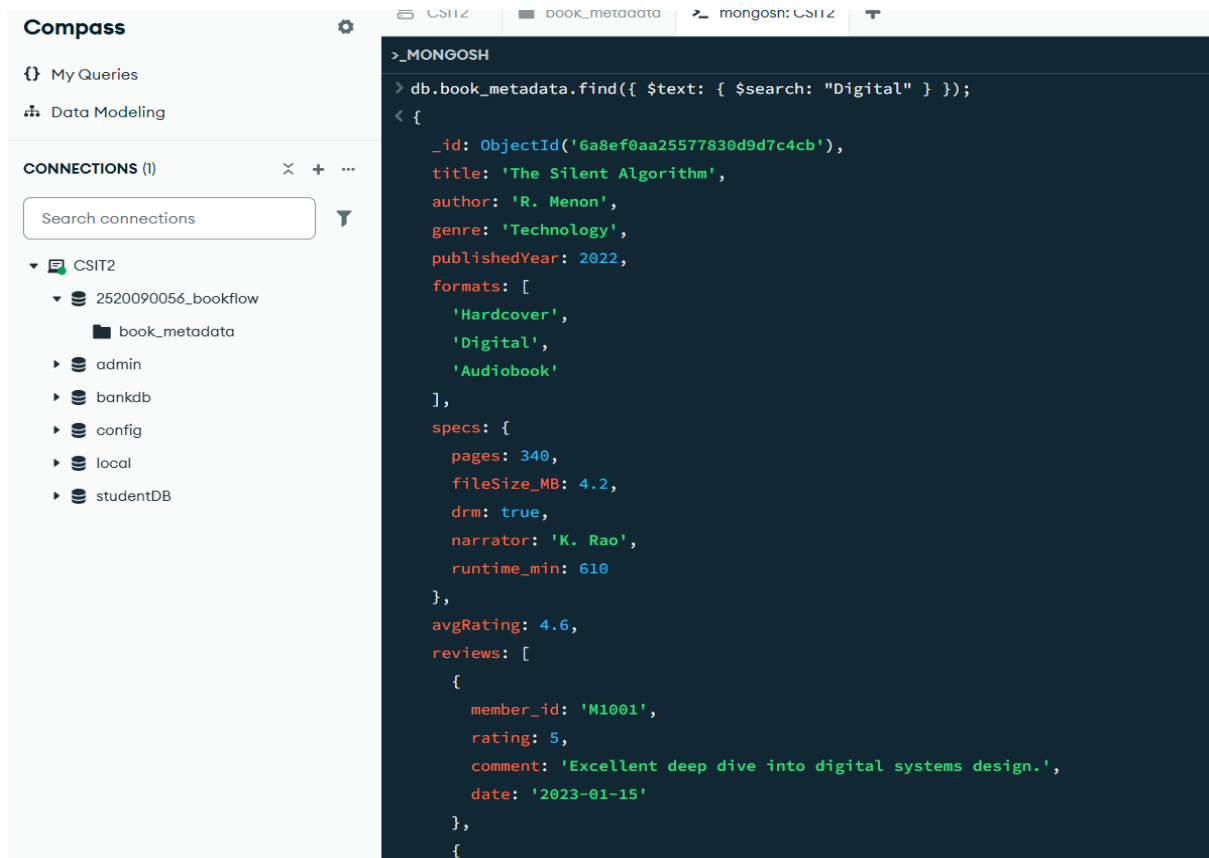
Task 4 : Indexing for Speed



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists a connection named 'CSIT2' with a database '2520090056_bookflow' and a collection 'book_metadata'. The main panel displays the MONGOSH console with the following commands and results:

```
>_MONGOSH
> db.book_metadata.createIndex({ "reviews.comment": "text" });
< reviews.comment_text
> db.book_metadata.find({ $text: { $search: "Digital" } });
< {
  _id: ObjectId('6a8ef0aa25577830d9d7c4cb'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: '2023-01-15'
    }
  ]
}
```

b) Query using the text index



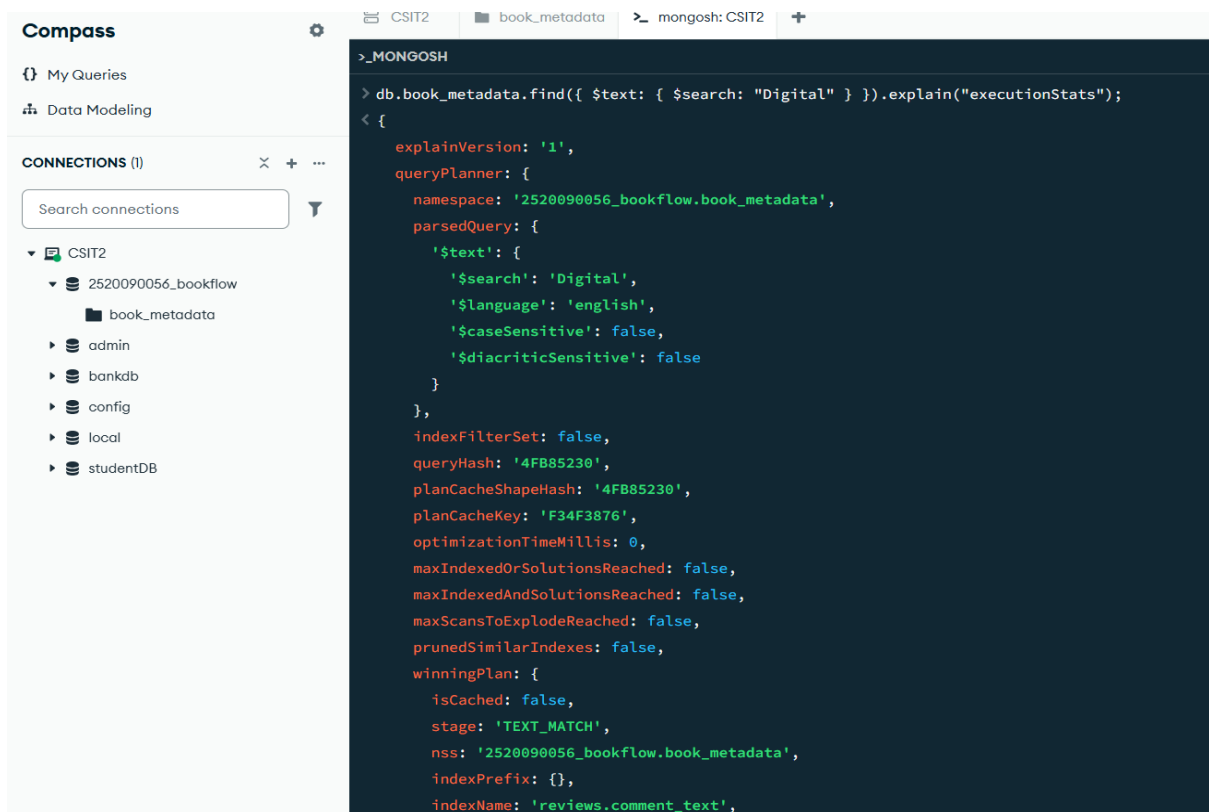
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'CSIT2' database, with the 'book_metadata' collection selected. The main panel displays a MongoDB query in the 'MONGOSH' tab:

```
> db.book_metadata.find({ $text: { $search: "Digital" } });
```

The result is a single document:

```
{
  "_id": ObjectId('6a8ef0aa25577830d9d7c4cb'),
  "title": 'The Silent Algorithm',
  "author": 'R. Menon',
  "genre": 'Technology',
  "publishedYear": 2022,
  "formats": [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  "specs": {
    "pages": 340,
    "fileSize_MB": 4.2,
    "drm": true,
    "narrator": 'K. Rao',
    "runtime_min": 610
  },
  "avgRating": 4.6,
  "reviews": [
    {
      "member_id": 'M1001',
      "rating": 5,
      "comment": 'Excellent deep dive into digital systems design.',
      "date": '2023-01-15'
    }
  ]
}
```

c) Verify index usage with .explain()



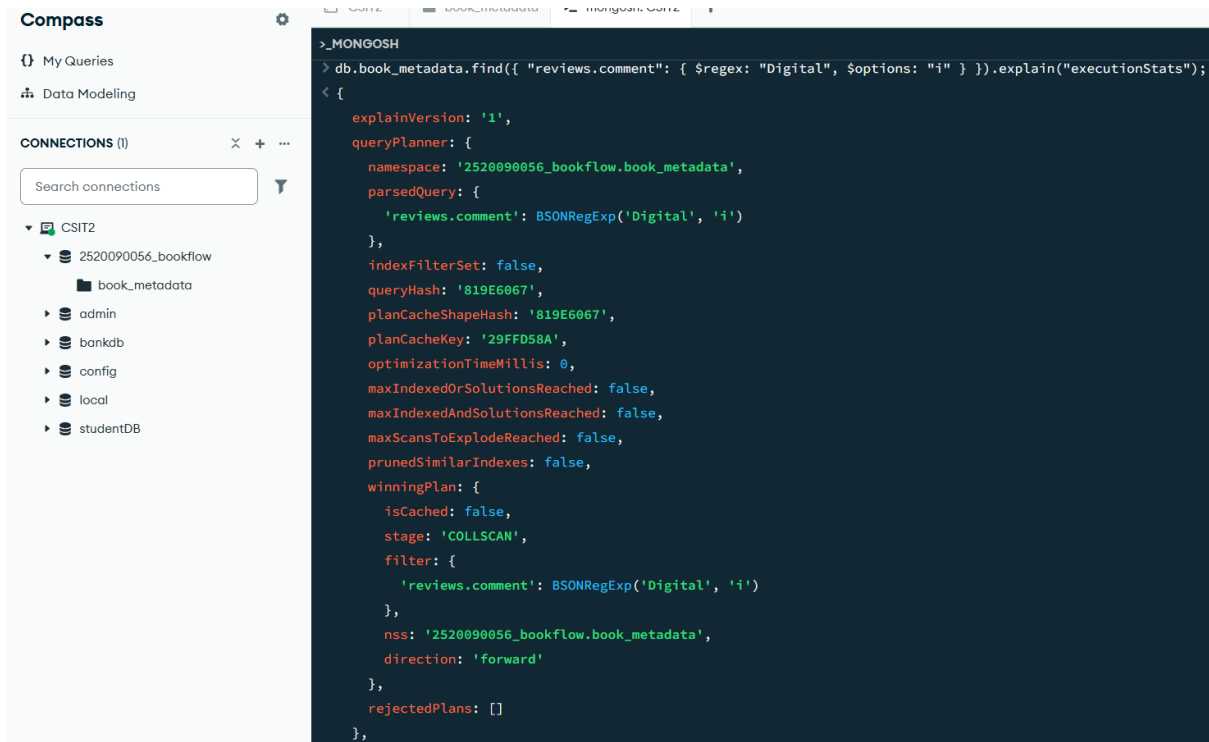
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'CSIT2' database, with the 'book_metadata' collection selected. The main panel displays a MongoDB query in the 'MONGOSH' tab:

```
> db.book_metadata.find({ $text: { $search: "Digital" } }).explain("executionStats");
```

The result is a document showing execution statistics:

```
{
  "explainVersion": '1',
  "queryPlanner": {
    "namespace": '2520090056_bookflow.book_metadata',
    "parsedQuery": {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    }
  },
  "indexFilterSet": false,
  "queryHash": '4FB85230',
  "planCacheShapeHash": '4FB85230',
  "planCacheKey": 'F34F3876',
  "optimizationTimeMillis": 0,
  "maxIndexedOrSolutionsReached": false,
  "maxIndexedAndSolutionsReached": false,
  "maxScansToExplodeReached": false,
  "prunedSimilarIndexes": false,
  "winningPlan": {
    "isCached": false,
    "stage": 'TEXT_MATCH',
    "nss": '2520090056_bookflow.book_metadata',
    "indexPrefix": {},
    "indexName": 'reviews.comment_text'
  }
}
```

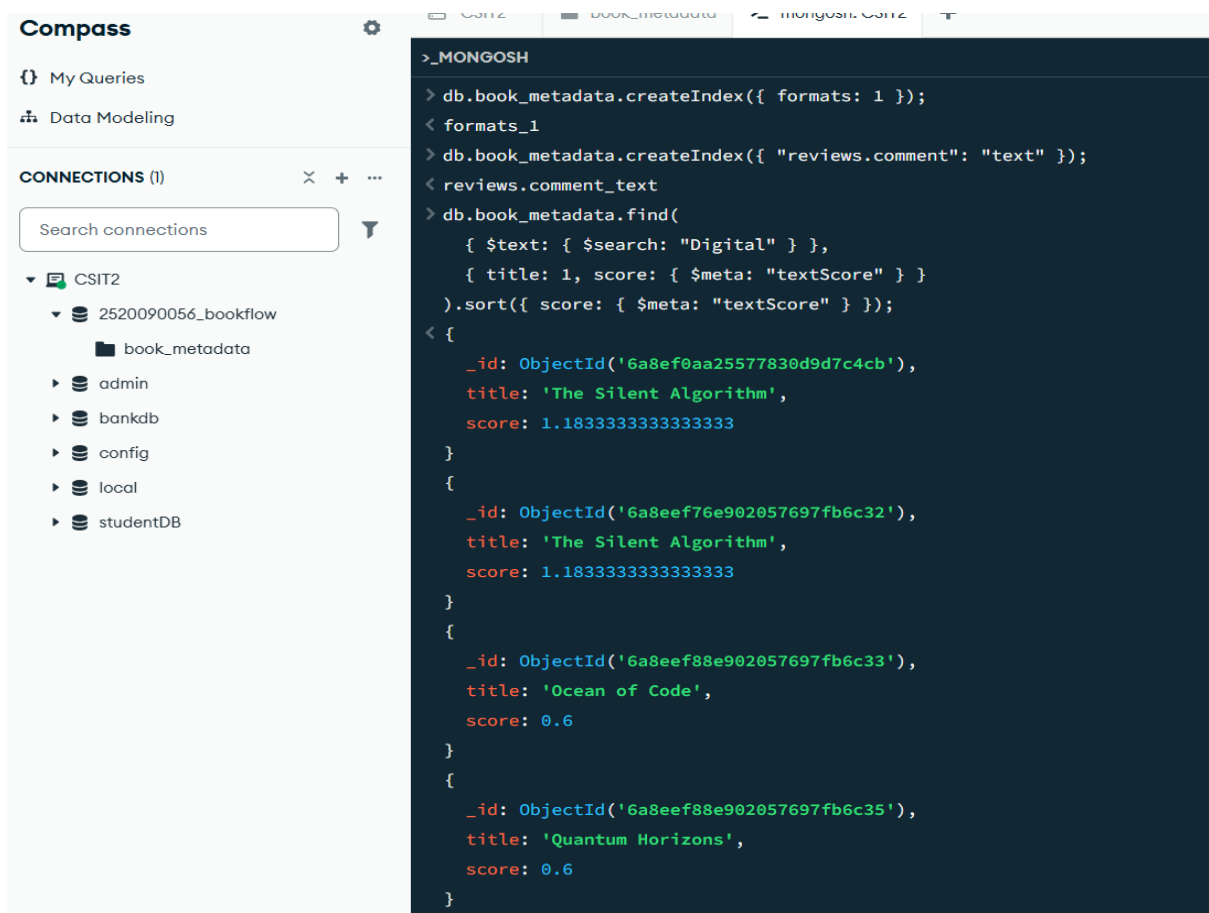
d) Compare against a non-indexed regex scan



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists a connection named 'CSIT2' with a database '2520090056_bookflow' and a collection 'book_metadata'. The main panel displays a MongoDB shell session with the following command and output:

```
>_MONGOSH
> db.book_metadata.find({ "reviews.comment": { $regex: "Digital", $options: "i" } }).explain("executionStats");
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: '2520090056_bookflow.book_metadata',
    parsedQuery: {
      'reviews.comment': BSONRegExp('Digital', 'i')
    },
    indexFilterSet: false,
    queryHash: '819E6067',
    planCacheShapeHash: '819E6067',
    planCacheKey: '29FFD58A',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: {
        'reviews.comment': BSONRegExp('Digital', 'i')
      },
      nss: '2520090056_bookflow.book_metadata',
      direction: 'forward'
    },
    rejectedPlans: []
  },
}
```

e) Supporting index for common queries (multikey index)



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists a connection named 'CSIT2' with a database '2520090056_bookflow' and a collection 'book_metadata'. The main panel displays a MongoDB shell session with the following commands and output:

```
>_MONGOSH
> db.book_metadata.createIndex({ formats: 1 });
< formats_1
> db.book_metadata.createIndex({ "reviews.comment": "text" });
< reviews.comment_text
> db.book_metadata.find(
  { $text: { $search: "Digital" } },
  { title: 1, score: { $meta: "textScore" } }
).sort({ score: { $meta: "textScore" } });
< {
  _id: ObjectId('6a8ef0aa25577830d9d7c4cb'),
  title: 'The Silent Algorithm',
  score: 1.1833333333333333
}
{
  _id: ObjectId('6a8ef76e902057697fb6c32'),
  title: 'The Silent Algorithm',
  score: 1.1833333333333333
}
{
  _id: ObjectId('6a8eef88e902057697fb6c33'),
  title: 'Ocean of Code',
  score: 0.6
}
{
  _id: ObjectId('6a8eef88e902057697fb6c35'),
  title: 'Quantum Horizons',
  score: 0.6
}
```

Compass

{ } My Queries

🔗 Data Modeling

CONNECTIONS (1)

Search connections

▼ CSIT2

- 2520090056_bookflow
 - book_metadata
- admin
- bankdb
- config
- local
- studentDB

>_MONGOSH

```
explainVersion: '1',
queryPlanner: {
  namespace: '2520090056_bookflow.book_metadata',
  parsedQuery: {
    '$text': {
      '$search': 'Digital',
      '$language': 'english',
      '$caseSensitive': false,
      '$diacriticSensitive': false
    }
  },
  indexFilterSet: false,
  queryHash: '4FB85230',
  planCacheShapeHash: '4FB85230',
  planCacheKey: 'F34F3876',
  optimizationTimeMillis: 0,
  maxIndexedOrSolutionsReached: false,
  maxIndexedAndSolutionsReached: false,
  maxScansToExplodeReached: false,
  prunedSimilarIndexes: false,
  winningPlan: {
    isCached: false,
    stage: 'TEXT_MATCH',
    nss: '2520090056_bookflow.book_metadata',
    indexPrefix: {},
    indexName: 'reviews.comment_text',
    parsedTextQuery: {
      terms: [
```

Compass

{ } My Queries

🔗 Data Modeling

CONNECTIONS (1)

Search connections

▼ CSIT2

- 2520090056_bookflow
 - book_metadata
- admin
- bankdb
- config
- local
- studentDB

>_MONGOSH

```
command: {
  find: 'book_metadata',
  filter: {
    '$text': {
      '$search': 'Digital'
    }
  },
  '$db': '2520090056_bookflow'
},
serverInfo: {
  host: 'LAPTOP-D2M8NKBH',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
```